

Online Java Compiler - online edit

https://www.onlinegdb.com/online_java_compiler#

Language: Java

Linear Search: Shoes
Binary Search: Shoes

...Program finished with exit code: 0
Press ENTER to exit console.

```
1 class Product {
2     int productId;
3     String productName;
4     String category;
5
6     Product(int productId, String productName, String category) {
7         this.productId = productId;
8         this.productName = productName;
9         this.category = category;
10    }
11 }
12
13 public class Main {
14
15     static Product linearSearch(Product[] products, int searchId) {
16         for (Product product : products) {
17             if (product.productId == searchId)
18                 return product;
19         }
20         return null;
21     }
22
23     static Product binarySearch(Product[] products, int searchId) {
24         int left = 0, right = products.length - 1;
25
26         while (left <= right) {
27             int mid = (left + right) / 2;
28
29             if (products[mid].productId == searchId)
30                 return products[mid];
31
32             if (products[mid].productId < searchId)
33                 left = mid + 1;
34             else
35                 right = mid - 1;
36         }
37         return null;
38     }
39
40     public static void main(String[] args) {
41
42         Product[] products = {
43             new Product(101, "Laptop", "Electronics"),
44             new Product(102, "Mobile", "Electronics"),
45             new Product(103, "Shoes", "Fashion"),
46             new Product(104, "Watch", "Accessories"),
47             new Product(105, "Book", "Education")
48         };
49
50         int searchId = 103;
51
52         Product result = linearSearch(products, searchId);
53         if (result != null)
54             System.out.println("Linear Search: " + result.productName);
55
56         result = binarySearch(products, searchId);
57         if (result != null)
58             System.out.println("Binary Search: " + result.productName);
59     }
60 }
```